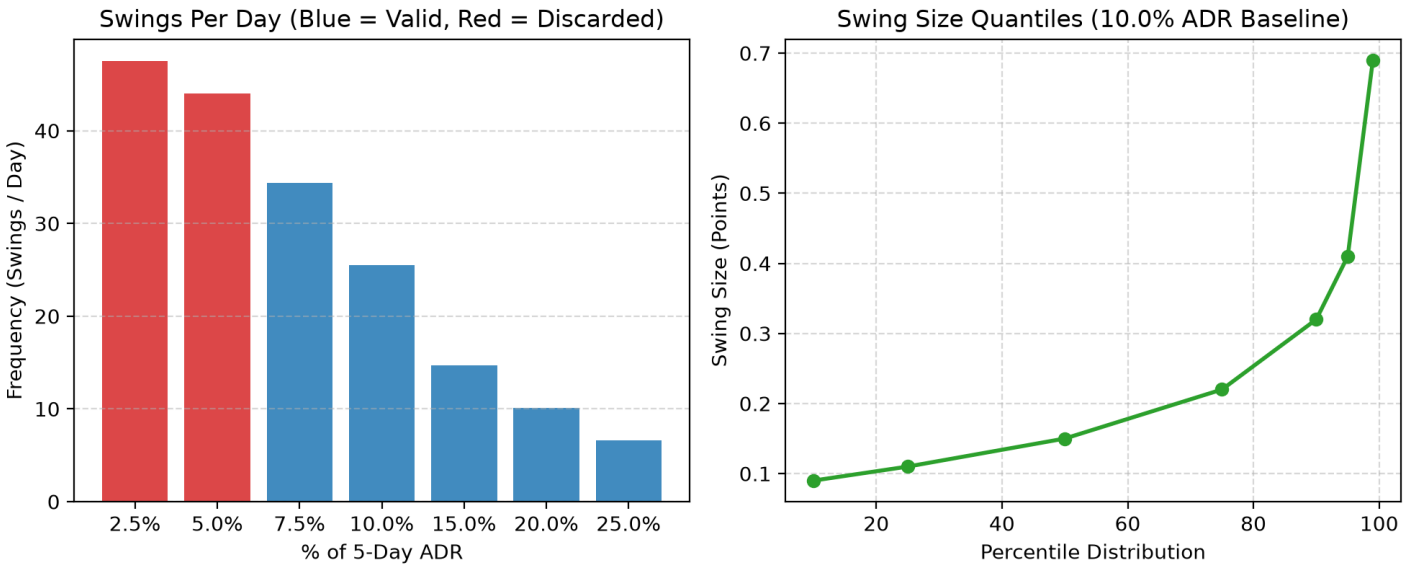


Statistical Swing Analysis Report: ITUB4 (5min)

5-Day Average Daily Range (ADR_5d): 0.74 pts

ITUB4 (5min) - Swing Distribution & Sensitivity Analysis



Parameter Sweep & Noise Filtering Matrix

Pct ADR	Threshold Pts	Total Swings	Swings Per Day	Avg Bars Per Swing	Mean Size	Median Size	P75 Size	P90 Size	P95 Size	Status
2.5%	0.02	23718	47.5	2.8	0.12	0.09	0.15	0.23	0.31	DISCARDED
5.0%	0.04	21943	44.0	3.0	0.13	0.1	0.16	0.24	0.32	DISCARDED
7.5%	0.06	17141	34.4	3.5	0.16	0.12	0.18	0.27	0.36	VALID
10.0%	0.07	12703	25.5	4.4	0.19	0.15	0.22	0.32	0.41	VALID
15.0%	0.11	7349	14.7	6.9	0.26	0.21	0.3	0.43	0.54	VALID
20.0%	0.15	5036	10.1	9.6	0.31	0.26	0.37	0.53	0.68	VALID
25.0%	0.19	3290	6.6	14.2	0.4	0.32	0.46	0.68	0.86	VALID